Module 7: Introduction to Gibbs Sampling

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Agenda

- Background knowledge (inverse CDF method)
- Gibbs sampling (two-stage sampler)
- Exponential example
- Normal example
- Pareto example
- Diagnostics

What will you learn in this lecture

- What is a Gibbs sampler?
- Some important properties of a Gibbs sampler
- What is needed in order to run a Gibbs sampler (the conditional distributions)
- How to find the conditional distributions for some examples
- Diagnostics used for monitoring "convergence" of the Gibbs sampler
- By the end of the lecture, you should know how to derive conditional distributions, run a Gibbs sampler, analyze diagnostics, and interpret summary statistics from the Gibbs sampler!

Background knowledge

The inverse CDF technique for generating a random sample uses the fact that a continuous CDF, F, is a one-to-one mapping of the domain of the CDF into the interval $(0, 1)$.

Lemma

If U is a uniform random variable on $(0, 1)$, then $X = F^{-1}(U)$ has the distribution F.

Formal Proof: https://www.youtube.com/watch?v=irheiVXJRm8

Gibbs sampler

- Suppose $p(x, y)$ is a p.d.f. or p.m.f. that is difficult to sample from directly.
- Suppose, though, that we *can* easily sample from the conditional distributions $p(x|y)$ and $p(y|x)$.
- The Gibbs sampler proceeds as follows:
  1. set x and y to some initial starting values
  2. then sample $x|y$, then sample $y|x$, then $x|y$, and so on.

Gibbs sampler

0. Set (x_0, y_0) to some starting value.

1. Sample $x_1 \sim p(x|y_0)$, that is, from the conditional distribution $X \mid Y = y_0$.
   Current state: (x_1, y_0)
   Sample $y_1 \sim p(y|x_1)$, that is, from the conditional distribution $Y \mid X = x_1$.
   Current state: (x_1, y_1)

2. Sample $x_2 \sim p(x|y_1)$, that is, from the conditional distribution $X \mid Y = y_1$.
   Current state: (x_2, y_1)
   Sample $y_2 \sim p(y|x_2)$, that is, from the conditional distribution $Y \mid X = x_2$.
   Current state: (x_2, y_2)
   $\vdots$

Repeat iterations 1 and 2, M times.

Gibbs sampler

This procedure defines a sequence of pairs of random variables

$$(X_0, Y_0), (X_1, Y_1), (X_2, Y_2), (X_3, Y_3), \ldots$$

Markov chain and dependence

$$(X_0, Y_0), (X_1, Y_1), (X_2, Y_2), (X_3, Y_3), \ldots$$

satisfies the property of being a Markov chain.

The conditional distribution of (X_{i+1}, Y_{i+1}) given all of the previous pairs depends only on (X_i, Y_i)

Example: The conditional distribution of (X_5, Y_5) given all of the previous pairs depends only on (X_4, Y_4)

$(X_0, Y_0), (X_1, Y_1), (X_2, Y_2), (X_3, Y_3), \ldots$ are not iid samples (Think about why).

Ideal Properties of MCMC

- (x_0, y_0) chosen to be in a region of high probability under $p(x, y)$, but often this is not so easy.
- We run the chain for M iterations and discard the first B samples $(X_1, Y_1), \ldots, (X_B, Y_B)$. This is called *burn-in*.
- Typically: if you run the chain long enough, the choice of B doesn't matter.
- Roughly speaking, the performance of an MCMC algorithm—that is, how quickly the sample averages $\frac{1}{N}\sum_{i=1}^{N} h(X_i, Y_i)$ converge—is referred to as the *mixing rate*.
- An algorithm with good performance is said to "have good mixing", or "mix well".

Gibbs is a type of MCMC

Under some conditions, assume that $E|(X, Y)| < \infty$. Assume that

$$(X, Y) \sim p(x, y)$$

Then

$$\frac{1}{M} \sum_{i=1}^{M} (X_i, Y_i) \to E[(X, Y)]$$

as $M \to \infty$.

Thus, we have an approximation of $E[(X, Y)]$, which is referred to MCMC in this setting (the pairs are not i.i.d.)

Exponential Example

Consider the following Exponential model for observation(s) $x = (x_1, \ldots, x_n)$.[1]:

$$p(x|a, b) = ab \exp(-abx) I(x > 0)$$

and suppose the prior is

$$p(a, b) = \exp(-a - b) I(a, b > 0).$$

You want to sample from the posterior $p(a, b|x)$.

[1] Please note that in the attached data there are 40 observations, which can be found in data-exponential.csv.

Conditional distributions

$$
\begin{aligned}
p(\boldsymbol{x}|a,b) &= \prod_{i=1}^{n} p(x_i|a,b) \\
&= \prod_{i=1}^{n} ab\exp(-abx_i) \\
&= (ab)^n \exp\left(-ab\sum_{i=1}^{n} x_i\right).
\end{aligned}
$$

The function is symmetric for a and b, so we only need to derive $p(a|\boldsymbol{x},b)$.

Conditional distributions

This conditional distribution satisfies

$$\begin{aligned} p(a|\boldsymbol{x}, b) &\propto_a p(a, b, \boldsymbol{x}) \\ &= p(\boldsymbol{x}|a, b)p(a, b) \\ &= \text{fill in full details for lab this week} \end{aligned}$$

Gibbs sampling code

```
knitr::opts_chunk$set(cache=TRUE)
library(MASS)
data <- read.csv("data-exponential.csv", header = FALSE)
```

Gibbs sampling code

```
########################################
# This function is a Gibbs sampler
#
# Args
#   start.a: initial value for a
#   start.b: initial value for b
#   n.sims: number of iterations to run
#   data:  observed data, should be in a
            # data frame with one column
#
# Returns:
#   A two column matrix with samples
     #   for a in first column and
# samples for b in second column
########################################
```

Gibbs sampling code

```
sampleGibbs <- function(start.a, start.b, n.sims, data){
  # get sum, which is sufficient statistic
  x <- sum(data)
  # get n
  n <- nrow(data)
  # create empty matrix, allocate memory for efficiency
  res <- matrix(NA, nrow = n.sims, ncol = 2)
  res[1,] <- c(start.a,start.b)
  for (i in 2:n.sims){
    # sample the values
    res[i,1] <- rgamma(1, shape = n+1,
                       rate = res[i-1,2]*x+1)
    res[i,2] <- rgamma(1, shape = n+1,
                       rate = res[i,1]*x+1)
  }
  return(res)
}
```

Gibbs sampler code

```
# run Gibbs sampler
n.sims <- 10000
# return the result (res)
res <- sampleGibbs(0.25,0.25,n.sims,data)
head(res)
```

```
##           [,1]      [,2]
## [1,] 0.250000 0.2500000
## [2,] 1.859753 0.1732889
## [3,] 1.998903 0.2278472
## [4,] 1.990581 0.2137003
## [5,] 2.558853 0.1887937
## [6,] 2.116031 0.1844649
```

Exponential Example

You will explore this problem more in lab this week and in your homework.

Toy Example

$$p(x, y) \propto e^{-xy} \mathbb{1}(x, y \in (0, c))$$

$$p(x|y) \underset{x}{\propto} p(x, y) \underset{x}{\propto} e^{-xy} \mathbb{1}(0 < x < c) \underset{x}{\propto} \text{Exp}(x|y) \mathbb{1}(x < c).^2$$

- $p(x|y)$ is a *truncated* version of the $\text{Exp}(y)$ distribution
- It is the same as taking $X \sim \text{Exp}(y)$ and conditioning on it being less than c, i.e., $X \mid X < c$.
- Let's refer to this as the $\text{TExp}(y, (0, c))$ distribution.

[2] Under $\propto$, we write the random variable (x) for clarity.

Toy Example

- The Gibbs sampling approach is to alternately sample from $p(x|y)$ and $p(y|x)$.
- Note $p(x, y)$ is symmetric with respect to x and y.
- Hence, only need to derive one of these and then we can get the other one by just swapping x and y.
- Let's look at $p(x|y)$.

Toy Example

An easy way to generate a sample from $Z \sim \text{TExp}(\theta, (0, c))$, is:

1. First, sample $U \sim \text{Uniform}(0, F(c|\theta))$ where

$$F(x|\theta) = 1 - e^{-\theta x}$$

   is the $\text{Exp}(\theta)$ c.d.f.

2. Set $Z = F^{-1}(U|\theta)$ where

$$F^{-1}(u|\theta) = -(1/\theta)\log(1-u)$$

   is the inverse c.d.f. for $u \in (0, 1)$.

There is an exercise on the next slide to prove step 2.

Toy Example

Practice Exercise: Verify that

$$F^{-1}(u|\theta) = -(1/\theta)\log(1-u).$$

Solution:

To solve for F^{-1}, set $u = F(x)$ for $u \in (0,1)$ and solve for x.

$$\begin{aligned} u &= 1 - e^{-\theta x} \implies \\ e^{-\theta x} &= 1 - u \implies \\ -\theta x &= \log(1-u) \implies \\ x &= -\frac{1}{\theta}\log(1-u). \end{aligned}$$

This proves that $F^{-1}(u|\theta) = -(1/\theta)\log(1-u)$.

Toy example

Let's apply Gibbs sampling, denoting $S = (0, c)$.

0. Initialize $x_0, y_0 \in S$.
1. Sample $x_1 \sim \mathrm{TExp}(y_0, S)$, then sample $y_1 \sim \mathrm{TExp}(x_1, S)$.
2. Sample $x_2 \sim \mathrm{TExp}(y_1, S)$, then sample $y_2 \sim \mathrm{TExp}(x_2, S)$.

   $\vdots$

N. Sample $x_N \sim \mathrm{TExp}(y_{N-1}, S)$, sample $y_N \sim \mathrm{TExp}(x_N, S)$.

Figure 1 demonstrates the algorithm, with $c = 2$ and initial point $(x_0, y_0) = (1, 1)$.

Toy example

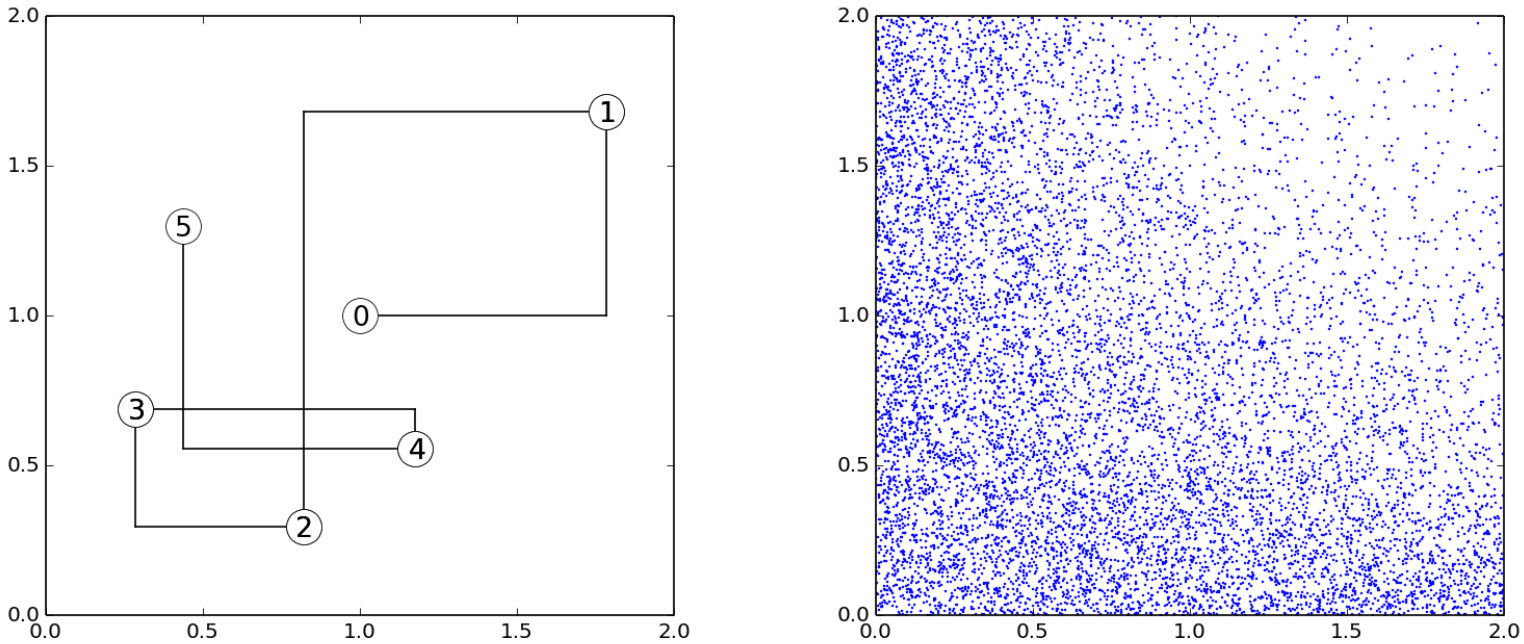


Figure 1: (Left) Schematic representation of the first 5 Gibbs sampling iterations/sweeps/scans. (Right) Scatterplot of samples from 10^4 Gibbs sampling iterations.

Example: Normal with semi-conjugate prior

Consider

$$X_1, \ldots, X_n | \mu, \lambda \overset{iid}{\sim} \mathcal{N}(\mu, \lambda^{-1}).$$

Then independently consider

$$\boldsymbol{\mu} \sim \mathcal{N}(\mu_0, \lambda_0^{-1})$$
$$\boldsymbol{\lambda} \sim \text{Gamma}(a, b)$$

We refer to this as a **semi-conjugate situation** for the following reasons:

1. The prior on μ is **conjugate** for each fixed value of λ since we get an updated Normal distribution.
2. The prior on λ is **conjugate** for each fixed value of μ since we get an updated Gamma distribution.

For ease of notation, denote the observed data points by $x_{1:n}$.

How does one derive $p(\mu, \lambda \mid x_{1:n})$?

Example

We know that for the Normal–Normal model, we know that for any fixed value of λ,

$$\boldsymbol{\mu}|\lambda, x_{1:n} \sim \mathcal{N}(M_\lambda, L_\lambda^{-1})$$

where

$$L_\lambda = \lambda_0 + n\lambda \text{ and } M_\lambda = \frac{\lambda_0\mu_0 + \lambda\sum_{i=1}^n x_i}{\lambda_0 + n\lambda}.$$

For any fixed value of μ, it is straightforward to derive[3] that

$$\boldsymbol{\lambda}|\mu, x_{1:n} \sim \text{Gamma}(A_\mu, B_\mu) \tag{1}$$

where $A_\mu = a + n/2$ and

$$B_\mu = b + \tfrac{1}{2}\sum(x_i - \mu)^2 = n\hat{\sigma}^2 + n(\bar{x} - \mu)^2$$

where $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$.

[3] derivation is found at the end of slides

Example

Goal: sample from $p(\mu, \lambda \mid x_{1:n})$

To implement Gibbs sampling in this example, each iteration consists of sampling:

$$\begin{aligned} \boldsymbol{\mu} | \lambda, x_{1:n} &\sim \mathcal{N}(M_\lambda, L_\lambda^{-1}) \\ \boldsymbol{\lambda} | \mu, x_{1:n} &\sim \text{Gamma}(A_\mu, B_\mu). \end{aligned}$$

This will give us samples

$$(\mu_0, \lambda_0), \ldots (\mu_S, \lambda_S)$$

Pareto example

Distributions of sizes and frequencies often tend to follow a "power law'' distribution.

- wealth of individuals
- size of oil reserves
- size of cities
- word frequency
- returns on stocks

Power law distribution

The power law (also called the scaling law) states that a relative change in one quantity results in a proportional relative change in another.

Example: One simple example to think of is a square. If we double the length of one side, from 2 to 4 inches, then the area will quadruple (from 4 to 16 inches squared).

A power law distribution has the form

$$Y = kX^{\alpha}$$

, where:

X, Y are random variables, k is a constant and α is a fixed exponent.

Power law distribution

The Pareto distribution with shape $\alpha > 0$ and scale $c > 0$ has p.d.f.

$$\text{Pareto}(x|\alpha, c) = \frac{\alpha c^\alpha}{x^{\alpha+1}} \mathbb{1}(x > c) \propto \frac{1}{x^{\alpha+1}} \mathbb{1}(x > c).$$

- This is referred to as a power law distribution, because the p.d.f. is proportional to x raised to a power.
- c is a lower bound on the observed values.
- We will use Gibbs sampling to perform inference for α and c.
- Let X be the population of a city.

Pareto example

Rank	City	Population
1	Charlotte	731424
2	Raleigh	403892
3	Greensboro	269666
4	Durham	228330
5	Winston-Salem	229618
6	Fayetteville	200564
7	Cary	135234
8	Wilmington	106476
9	High Point	104371
10	Greenville	84554
11	Asheville	85712
12	Concord	79066
⋮	⋮	⋮
44	Havelock	20735
45	Carrboro	19582
46	Shelby	20323
47	Clemmons	18627
48	Lexington	18931
49	Elizabeth City	18683

Parameter intepretations

- α tells us the scaling relationship between the size of cities and their probability of occurring.
  - Let $\alpha = 1$.
  - Density looks like $1/x^{\alpha+1} = 1/x^2$.
  - Cities with 10,000–20,000 inhabitants occur roughly $10^{\alpha+1} = 100$ times as frequently as cities with 100,000–110,000 inhabitants (or $10^{\alpha+1}/10 = 10$ times as frequently as cities with 100,000-200,000 inhabitants)
- c represents the cutoff point—any cities smaller than this were not included in the dataset.

Prior selection

For simplicity, let's use an **(improper) default prior**:

$$p(\alpha, c) \propto \mathbb{1}(\alpha, c > 0).$$

Recall:

- An *improper/default prior* is a non-negative function of the parameters which integrates to infinity.
- Often (but not always!) the resulting "posterior" will be proper.
- It is important that the "posterior" be proper, since otherwise the whole Bayesian framework breaks down.

Pareto example

Recall

$$p(x|\alpha, c) = \frac{\alpha c^\alpha}{x^{\alpha+1}} \mathbb{1}(x > c) \tag{2}$$

$$\mathbb{1}(\alpha, c > 0) \tag{3}$$

Let's derive the posterior:

$$\begin{aligned} p(\alpha, c|x_{1:n}) &\overset{\text{def}}{\underset{\alpha,c}{\propto}} p(x_{1:n}|\alpha, c)p(\alpha, c) \\ &\underset{\alpha,c}{\propto} \mathbb{1}(\alpha, c > 0) \prod_{i=1}^{n} \frac{\alpha c^\alpha}{x_i^{\alpha+1}} \mathbb{1}(x_i > c) \\ &= \frac{\alpha^n c^{n\alpha}}{(\prod x_i)^{\alpha+1}} \mathbb{1}(c < x_*)\mathbb{1}(\alpha, c > 0) \end{aligned} \tag{4}$$

where $x_* = \min\{x_1, \ldots, x_n\}$.

Pareto example

As a joint distribution on (α, c),

- this does not seem to have a recognizable form,
- and it is not clear how we might sample from it directly.

What it needed to set up a Gibbs sampler?

Gibbs sampling

Let's try Gibbs sampling! To use Gibbs, we need to be able to sample $\alpha|c, x_{1:n}$ and $c|\alpha, x_{1:n}$.

By Equation 4, we find that

$$\begin{aligned} p(\alpha|c, x_{1:n}) &\underset{\alpha}{\propto} p(\alpha, c|x_{1:n}) \underset{\alpha}{\propto} \frac{\alpha^n c^{n\alpha}}{(\prod x_i)^\alpha} \mathbb{1}(\alpha > 0) \\ &= \alpha^n \exp\big(-\alpha(\textstyle\sum \log x_i - n \log c)\big) \mathbb{1}(\alpha > 0) \\ &\underset{\alpha}{\propto} \text{Gamma}\left(\alpha \,\middle|\, n+1, \textstyle\sum \log x_i - n \log c\right), \end{aligned}$$

and

$$p(c|\alpha, x_{1:n}) \underset{c}{\propto} p(\alpha, c|x_{1:n}) \underset{c}{\propto} c^{n\alpha} \mathbb{1}(0 < c < x_*),$$

which we will define to be $\text{Mono}(n\alpha + 1, x_*)$, and we define generally on the next slide.

Mono distribution

Here, we define the Mono distribution generally before returning to our example.

For $a > 0$ and $b > 0$, define the distribution Mono(a, b) (for monomial) with p.d.f.

$$\text{Mono}(x|a, b) \propto x^{a-1}\mathbb{1}(0 < x < b).$$

Since $\int_0^b x^{a-1}dx = b^a/a$, we have

$$\text{Mono}(x|a, b) = \frac{a}{b^a}x^{a-1}\mathbb{1}(0 < x < b),$$

and for $0 < x < b$, the c.d.f. is

$$F(x|a, b) = \int_0^x \text{Mono}(y|a, b)dy = \frac{a}{b^a}\frac{x^a}{a} = \frac{x^a}{b^a}.$$

Pareto example

To use the inverse c.d.f. technique, we solve for the inverse of F on $0 < x < b$: Let $u = \frac{x^a}{b^a}$ and solve for x.

$$u = \frac{x^a}{b^a} \tag{5}$$

$$b^a u = x^a \tag{6}$$

$$b u^{1/a} = x \tag{7}$$

Can sample from Mono(a, b) by drawing $U \sim$ Uniform$(0, 1)$ and setting $X = bU^{1/a}$.[4]

[4] It turns out that this is an inverse of the Pareto distribution, in the sense that if $X \sim$ Pareto(α, c) then $1/X \sim$ Mono$(\alpha, 1/c)$.

Pareto example

So, in order to use the Gibbs sampling algorithm to sample from the posterior $p(\alpha, c|x_{1:n})$, we initialize α and c, and then alternately update them by sampling:

$$\begin{aligned} \alpha|c, x_{1:n} &\sim \text{Gamma}\left(n+1, \sum \log x_i - n \log c\right) \\ c|\alpha, x_{1:n} &\sim \text{Mono}(n\alpha + 1, x_*). \end{aligned}$$

Traceplots

Traceplots. A traceplot simply shows the sequence of samples, for instance $\alpha_1, \ldots, \alpha_N$, or $c_1, \ldots, c_N$. Traceplots are a simple but very useful way to visualize how the sampler is behaving.

Traceplots

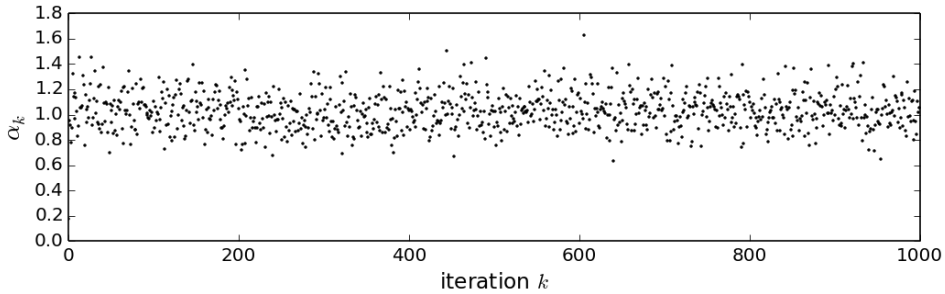


Figure 2: Traceplot of α

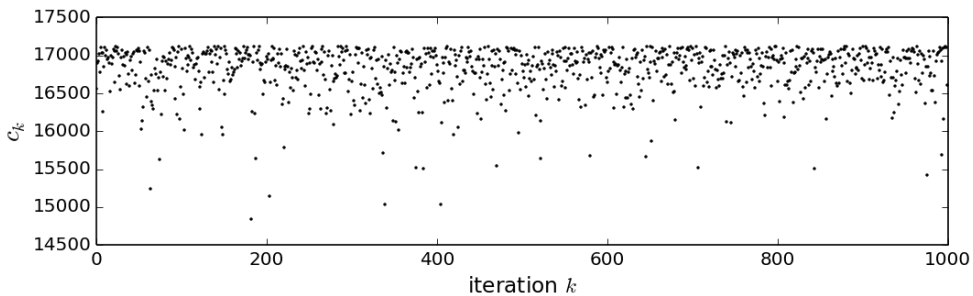


Figure 3: Traceplot of c.

Estimated density

Estimated density. We are primarily interested in the posterior on α, since it tells us the scaling relationship between the size of cities and their probability of occurring.

By making a histogram of the samples $\alpha_1, \ldots, \alpha_N$, we can estimate the posterior density $p(\alpha|x_{1:n})$.

The two vertical lines indicate the lower ℓ and upper u boundaries of an (approximate) 90% credible interval $[\ell, u]$—that is, an interval that contains 90% of the posterior probability:

$$\mathbb{P}(\boldsymbol{\alpha} \in [\ell, u] | x_{1:n}) = 0.9.$$

Estimated density

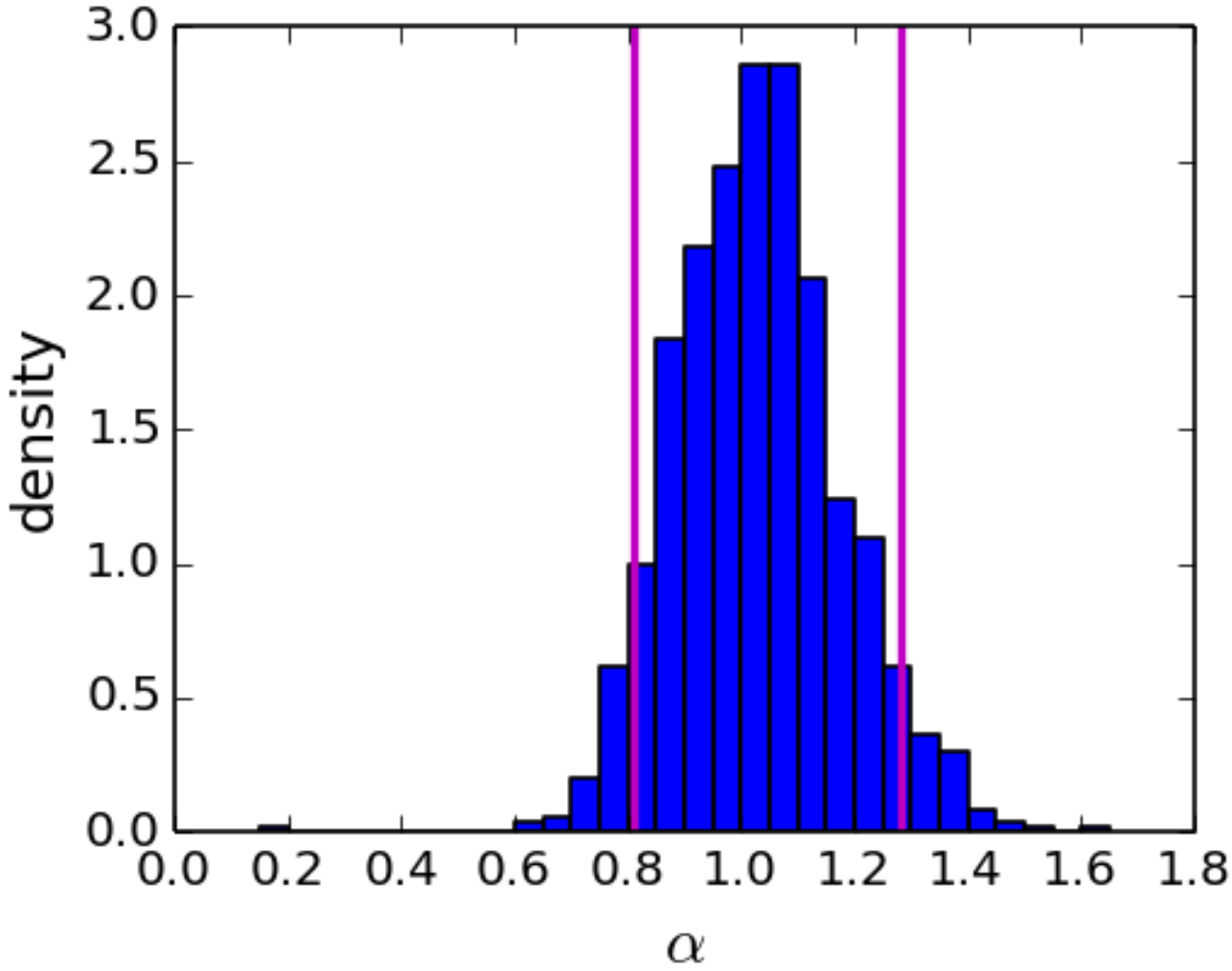


Figure 4: Estimated density of $\alpha|x_{1:n}$ with $\approx$ 90 percent credible intervals.

Running averages

Running averages. Panel (d) shows the running average $\frac{1}{k}\sum_{i=1}^{k}\alpha_i$ for $k = 1, \ldots, N$.

In addition to traceplots, running averages such as this are a useful heuristic for visually assessing the convergence of the Markov chain.

The running average shown in this example still seems to be meandering about a bit, suggesting that the sampler needs to be run longer (but this would depend on the level of accuracy desired).

Running averages

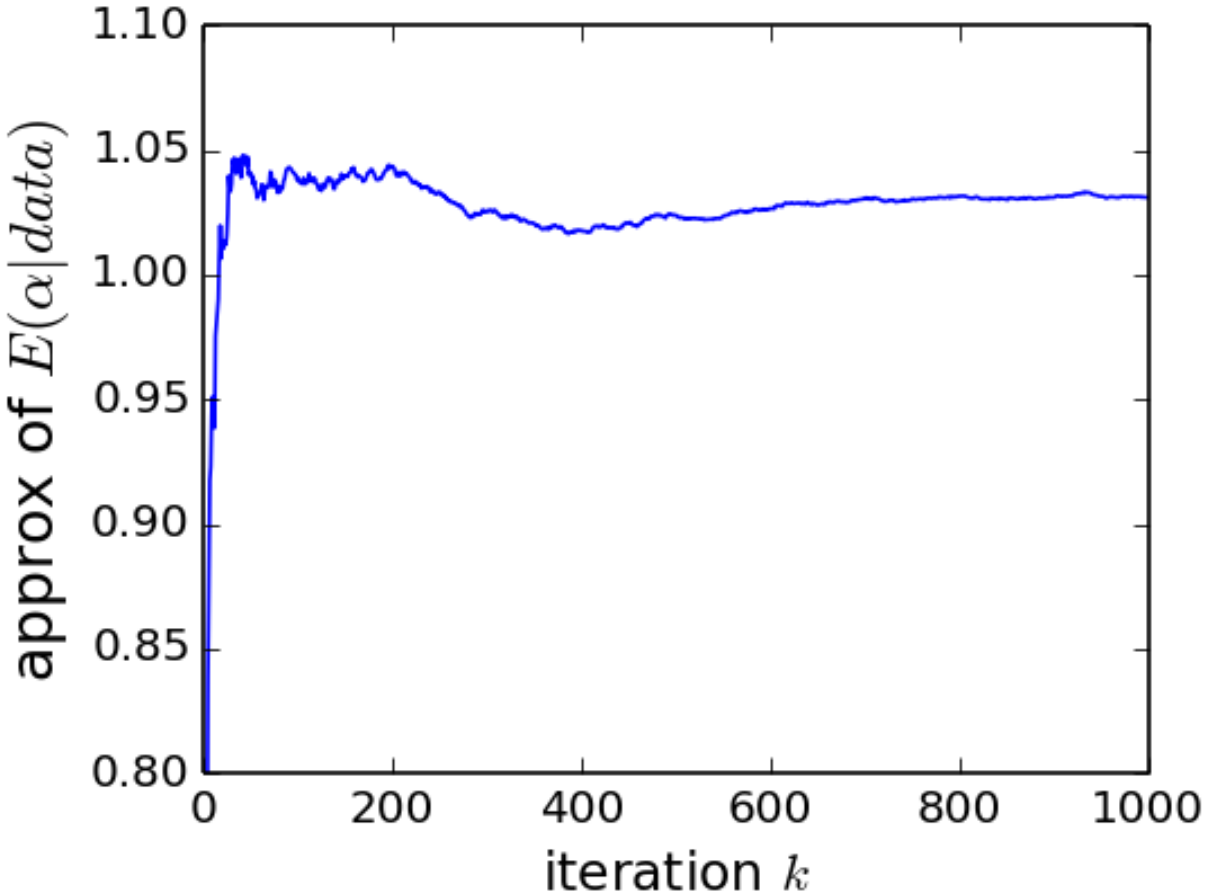


Figure 5: Running average plot

Detailed Takeways

- inverse CDF method
- Two-stage Gibbs sampler
- Markov Chain
- properties of Markov chains
- Exponential Example
- Truncated Exponential
- Normal-Normal-Gamma
- Pareto Case Study
- Trace plots
- Estimated Densities
- Running Aveage Plots

In class notes

Notes on burn-in can be found here:
https://github.com/resteorts/modern-bayes/blob/master/lecturesModernBayes20/lecture-7/class-notes/gibbs-partI/burn-in.pdf

Notes on Exponential example can be found here:
https://github.com/resteorts/modern-bayes/blob/master/lecturesModernBayes20/lecture-7/class-notes/gibbs-partI/gibbs-exponential-example.pdf

Notes on truncated exponential example can be found here:
https://github.com/resteorts/modern-bayes/blob/master/lecturesModernBayes20/lecture-7/class-notes/gibbs-partI/gibbs-truncated-exponential-example.pdf

Notes on the two-stage Gibbs sampler set up can be found here:
https://github.com/resteorts/modern-bayes/blob/master/lecturesModernBayes20/lecture-7/class-notes/gibbs-partI/intro-gibbs.pdf

Exercise

Recall that

$$X_1, \ldots, X_n \mid \lambda \overset{iid}{\sim} \text{Normal}(\mu, \lambda^{-1}) \tag{8}$$

$$\lambda \sim \text{Gamma}(a, b) \tag{9}$$

Let's derive the posterior update of $\lambda \mid x_{1:n}$, which we utilized in an earlier derivation.

Exercise

$$p(\lambda \mid x_{1:n}) \propto p(x_{1:n} \mid \lambda)p(\lambda) \tag{10}$$

$$= \prod_{i=1}^{n}[\sqrt{\frac{\lambda}{2\pi}}\exp\{-\frac{\lambda}{2}(x_i-\mu)^2\}] \times \frac{b^a}{\Gamma(a)}\lambda^{a-1}e^{-b\lambda} \tag{11}$$

$$\propto (\lambda)^{n/2}\exp\{-\frac{\lambda}{2}\sum_{i=1}^{n}(x_i-\mu)^2\}] \times \lambda^{a-1}e^{-b\lambda} \tag{12}$$

$$\propto (\lambda)^{a+n/2-1}\exp\{\lambda[b+\sum_{i=1}^{n}(x_i-\mu)^2]\} \tag{13}$$

This implies that

$$\lambda \mid x_{1:n} \sim \text{Gamma}(a+n/2, b+\sum_{i=1}^{n}(x_i-\mu)^2) \tag{14}$$